

CANOPY STRUCTURE & MOONLIGHT PIPELINE

Purpose: Build a single analysis dataset containing:

- Astronomical variables (Moonlit)
- LiDAR-derived canopy structure
- CoveR canopy photograph metrics

Output: master_log_lidar_canopy

Packages

```
library(dplyr)
library(sf)
library(terra)
install.packages("moonlit")
library(moonlit)
library(purrr)
library(lubridate)
```

```
master_log <- Scoring_master_log_Master_log
```

- Load the master log
- this should contain: Site, elevation, latitude, longitude, full_ID, stationID, filename, date, time, time_stamp, PESO, SOREX, UNK, VOLE, SCIURIDAE, MUSTELIDAE, OTHER

SECTION A: Moonlight (Moonlit package)

```
master_log_moon <- master_log |>
```

```
  mutate(
    time_stamp = ymd_hms(
      time_stamp,
      tz = "America/Los_Angeles"
    )
  )
```

- Convert timestamp column, convert to POSIXct (**R's standard date-time class**) with local timezone (have to do this for calculateMoonlightIntensity function
 - For example, if your timestamps are stored as text: "2026-06-19 22:15:03" R sees that as just a character string.
 - After conversion: as.POSIXct("2026-06-19 22:15:03", tz = "America/Los_Angeles") R understands that it represents:

- **Date:** June 19, 2026
- **Time:** 10:15:03 PM
- **Time zone:** Pacific Time ([America/Los_Angeles](#))

```
moon_data <- map_dfr(
  seq_len(nrow(master_log_moon)),
  ~ calculateMoonlightIntensity(
    lat = master_log_moon$latitude[.x],
    lon = master_log_moon$longitude[.x],
    date = master_log_moon$time_stamp[.x],
    e = 0.28
  )
)
```

- `map_dfr()` is used because `calculateMoonlightIntensity()` only calculates the moonlight values for one location and one time at a time. Our dataset has many rows, each with a different latitude, longitude, and timestamp, so you have to run the function once for every row and then combine all the results! Comes from `Purrr` package
- **map** = iterate over each element (similar to `lapply()`)
- **df** = the output is a data frame
- **r** = combine the results by **rows**

```
moon_data <- moon_data |>
  select(
    night,
    sunAltDegrees,
    moonAltDegrees,
    moonlightModel,
    twilightModel,
    illumination,
    moonPhase
  )
```

- Selecting the output columns you want

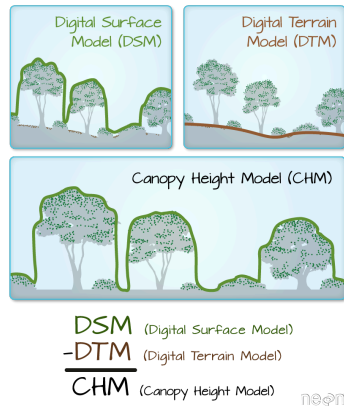
```
master_log_moon <- bind_cols(
  master_log_moon,
  moon_data
)
```

- Binding the moon data with the master log

SECTION B: LiDAR Canopy Structure

LiDAR provides three-dimensional measurements of vegetation. This section calculates canopy height surrounding each camera station using a **30 m** buffer.

- This buffer can be changed, consider using a 10m buffer instead
- We are using a 5m radius so 10m **diameter** for our plant data so a 10m buffer (20m diameter) may be best. Going smaller than that can be difficult with lidar.



We need to get: **Digital Terrain Model (or DTM)**: ground elevation or the elevation of the Earth's surface (sometimes also called a DEM or digital elevation model).

Digital Surface Model (or DSM): top of the surface (imagine draping a sheet over the canopy of a forest)

Canopy Height Model (CHM): The height of objects above the ground.

```
dem <-
```

```
rast("/Users/whitnymaxfield/Downloads/LDQ-45122D6/2014_OLC_Metro/Bare_Earth/bh45122d6")
```

- Loading the Bare Earth or ground surface LiDAR data (the DEM)

```
dsm <-
```

```
rast("/Users/whitnymaxfield/Downloads/LDQ-45122D6/2014_OLC_Metro/Highest_Hit/hh45122d6")
```

- Loading the Highest point or top of vegetation LiDAR data (the DSM)

```
chm <- dsm - dem
```

- The CHM is the difference of DSM and the DEM. Simply subtract them from each other to get the CHM.

```
chm[chm < 0] <- 0
```

- Occasionally small negative values occur because of raster misalignment or interpolation error. Since vegetation cannot have negative height, set these values to zero! This avoids errors later on.

```
buckets <- master_log |>
```

```
group_by(stationID) |>
```

```
summarise(
```

```
latitude = first(latitude),
```

```
longitude = first(longitude)
```

)

- Create one point per camera station
- The master log contains many observations for each station.
- But we only need one location per station for extracting habitat variables! (kind of like the bucket trap glossary on the first page of the camera trap data spreadsheet)
 - You should always check this output because sometimes small errors in lat lon can create duplicates which will cause issues.

```
bucket_sf <- buckets |>
```

```
  st_as_sf(  
    coords = c("longitude", "latitude"),  
    crs = 4326
```

```
  ) |>
```

```
  st_transform(crs(chm))
```

- taking the regular data frame of bucket locations and turning it into a spatial object that can be used with the canopy height model (CHM).
- Also changes the bucket locations into the same coordinate system as the canopy height model (CHM).

```
bucket_sf_filtered <- bucket_sf |>
```

```
  filter(!stationID %in% c("GOME01", "GOME02"))
```

- Remove stations outside LiDAR coverage. Since this pipeline ONLY works on the Tryon locations (have to use a different downloaded lidar set for Sauvie) we need to filter out the Sauvie locations like GOME, WARO, and BOBS

```
buffers <- st_buffer(bucket_sf_filtered, 30)
```

- Create **30** m habitat buffers (can be changed)!

```
vals <- terra::extract(  
  chm,
```

```
  vect(buffers),  
  ID = TRUE
```

```
)
```

- Extract canopy height values. Essentially asking "Which raster values are located inside these polygons?" In this case:
 - **Raster:** the canopy height model (**chm**)
 - **Polygons:** the buffers around each bucket
 - So we are extracting all canopy heights within each buffer

```
vals$stationID <- bucket_sf_filtered$stationID[vals$ID]
```

```
names(vals)[2] <- "chm_height"
```

- Match extraction IDs back to station names, rename raster column for clarity

```
canopy_metrics <- vals |>
```

```
group_by(stationID) |>
summarise(

  mean_height = mean(chm_height, na.rm = TRUE),

  max_height = max(chm_height, na.rm = TRUE),

  sd_height = sd(chm_height, na.rm = TRUE),

  cover2m = mean(chm_height > 2, na.rm = TRUE)

)
```

- Summarize canopy model

```
analysis_df <- master_log_moon |>
left_join(
  canopy_metrics,
  by = "stationID"
)
```

- Add LiDAR metrics to the observation dataset (add what we just calculated back into our working master log!)

SECTION C: Canopy Photographs (CoveR)

Canopy photographs provide fine-scale canopy structures that complement the LiDAR-derived model. Only **upper** canopy photographs from Tryon Creek stations are used in this workflow.

- The canopy photo analysis is performed separately using the CoveR pipeline!!!
- RUN COVER PIPELINE FIRST THEN UPLOAD RESULTS AS CANOPY_COVER_RESULTS OR RENAME TO MATCH

```
canopy_upper <- Canopy_Cover_Results |>
filter(
  grepl("^BOHO|^SOWE", Site),
  grepl("Upper", Image)
) |>
select(
  Site,

  CC, # Crown Cover

  FC, # Foliage Cover

  CP, # Crown Porosity
```

Le, # Effective Leaf Area Index

L, # Leaf Area Index

CI # Clumping Index

)

- filtering for Tryon sites and Upper images ONLY
- select variables wanted

SECTION D: Final Analysis Dataset

```
master_log_lidar_canopy <- analysis_df |>
```

```
left_join(
```

```
  canopy_upper,
```

```
  by = c("stationID" = "Site")
```

```
)
```

- Merge:
 - Camera observations
 - Moonlit variables
 - LiDAR canopy metrics
 - CoveR canopy metrics
- Final output: master_log_lidar_canopy
 - Each row represents one camera observation with both observation-level (moondata) and station-level (lidar and canopy) environmental variables.

```
write.csv(
```

```
  master_log_lidar_canopy,
```

```
  "LidarCanopyMoon_masterLog.csv",
```

```
  row.names = FALSE
```

```
)
```

- Turn it into a CSV

Helpful sites and information:

https://tgoodbody.github.io/lidRtutorial/04_chm.html

- I don't use the lidr package but still an interesting read

<https://gis.dogami.oregon.gov/maps/lidarviewer/>

- Where you download the data

<https://lab.jonesctr.org/lidar-processing-for-forestry-and-natural-resources-module-1-viewing-data/>

- Also lots of super helpful information and potentially some good future methods here
https://tgoodbody.github.io/lidRtutorial/01_read.html
- LidR package info, don't use in current pipeline
<https://earthdatascience.org/courses/earth-analytics/lidar-raster-data-r/lidar-chm-dem-dsm/>
- Super helpful
<https://rspatial.org/spatial/index.html>
- Downloaded lidar from here. This is a HUGE data set. mega massive like 4GB download at your own risk lmao
<https://gitlab.com/fchianucci/coveR>
- coveR page
<https://github.com/msmielak/moonlit>
- Moonlit page